Dynamics-Aware Gradient Clipping: Separating Feedback from Threshold Scheduling

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Abstract

Adaptive gradient clipping can improve training without establishing that its feedback signals are useful. We examine Dynamics-Aware Gradient Clipping (DAGC), a bounded threshold controller driven by clipping exposure and consecutive-gradient alignment. Its exposure term tightens clipping when clipping is already frequent. We show that an interior target-exposure equilibrium is unstable in a stationary mean-field model, even with exponential smoothing. We then compare the frozen controller with tuned clipping rules, an alignment ablation, and a predetermined threshold schedule on FashionMNIST and CIFAR-10, using validation-only calibration and twelve paired confirmation seeds in each of three conditions. Across 405 audited runs, DAGC falls below validation-tuned fixed clipping by 0.18, 1.17, and 0.29 percentage points in the three confirmation conditions. All three differences favor fixed clipping after multiplicity correction. DAGC and calibration-trace replay meet the prespecified, per-comparison ± 0.5 point equivalence criterion in all three conditions. The controller occupies its lower bound on 97.2–99.1% of updates, and removing alignment leaves logged CIFAR-10 trajectories unchanged. These controls separate threshold scheduling from evidence for an online feedback mechanism; the analysis does not claim convergence of the coupled training dynamics.

1. Introduction

Gradient clipping limits the contribution of a large gradient to an optimizer update (Pascanu et al., 2013). A fixed threshold is simple but introduces a scale-sensitive hyperparameter. Adaptive rules can instead derive a threshold from gradient history or parameter magnitude (Seetharaman et al., 2020; Brock et al., 2021). When an adaptive method succeeds, however, its accuracy alone does not establish why: a useful threshold profile, a smaller effective step size, and informative feedback can produce similar outcomes.

We examine that ambiguity in Dynamics-Aware Gradient Clipping (DAGC). The controller uses the recent frequency of clipping and the alignment of consecutive raw gradients. Its first version, V1, allowed exposure alone to increase the threshold but not decrease it. V2 replaces that one-sided term with a term that becomes negative above a reference exposure. This makes sustained clipping more restrictive. It also gives the feedback the opposite sign to a clipping-rate tracker: lowering a threshold makes further clipping more likely.

This distinction changes both the interpretation and the experiment needed to support the method. A comparison with an arbitrary fixed threshold cannot establish that exposure or alignment provides useful information beyond the gradient scale. Removing the alignment term tests its necessity. A lagged median rule tests whether magnitude information is sufficient. Replaying a threshold trace constructed on separate calibration runs tests the incremental value of adapting that threshold to the new training trajectory. Replay still uses the current norm to apply clipping; only its threshold is predetermined.

Our contribution is an analysis and controlled evaluation of this concrete controller. First, we derive its local positive-feedback behavior with the exposure EMA included. Second, we define a reproducible comparison with validation-only parameter selection, identical paired training streams, and explicit failure accounting. Third, we preserve the earlier negative mechanism controls and evaluate transfer to longer training and a residual network on RGB images. The scope is small image classifiers trained with momentum SGD; there is no claim of broad optimizer, architecture, or language-model transfer.

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2. Related Work

Adaptive thresholds. AutoClip selects a percentile of the observed gradient-norm history (Seetharaman et al., 2020). AGC bounds each output unit's gradient norm relative to its parameter norm (Brock et al., 2021). These supply magnitude-based comparators with different invariances and granularities. The median control in our study is simpler: it multiplies a lagged rolling median of global norms by a fixed coefficient. It is motivated by the moving bound already present inside DAGC, rather than proposed as a new clipping algorithm.

Rate feedback and clipping theory. Adaptive clipping for private federated learning estimates a desired norm quantile online (Andrew et al., 2021). Expressed in terms of clipping frequency, excess clipping increases its threshold. DAGC-V2 uses the opposite sign; it is neither a quantile estimator nor the first multiplicative clipping-rate controller. Analyses of clipping under relaxed smoothness (Zhang et al., 2020) and of stochastic clipping bias (Koloskova et al., 2023) show why clipping can affect optimization, but do not establish convergence of DAGC. Our proposition addresses a local controller model with an exogenous norm distribution, not the full learning process.

Evidence for a mechanism. Variation in initialization and hyperparameter selection can materially change benchmark conclusions (Bouthillier et al., 2021). We treat a training seed as the replication unit and report differences within paired seeds. The official test set is withheld from the new campaign's tuning. Earlier FashionMNIST test scores were used during method development, so this retrospective correction does not make the project's historical test exposure disappear. We distinguish the new fixed protocol from a completely untouched external benchmark.

3. Controller and Competing Explanations

Let $g_t = \nabla_\theta \ell_t(\theta_t)$ be the raw minibatch gradient, and let $n_t = \|g_t\|_2$. A positive threshold c_t gives

$$\widetilde{g}_t = \alpha_t g_t, \qquad \alpha_t = \min\{1, c_t / \max(n_t, \epsilon)\}. \tag{1}$$

We log exposure $e_t = \mathbf{1}[n_t > c_t]$, intensity $I_t = 1 - \alpha_t$, and alignment $a_t = \langle g_t, g_{t-1}\rangle / \max(n_t n_{t-1}, \epsilon)$. The first alignment is one. The exposure and alignment states are

$$E_t = \beta E_{t-1} + (1-\beta) e_t, \tag{2}$$
$$A_t = \beta_a A_{t-1} + (1-\beta_a) a_t, \tag{3}$$

initialized at $E_{-1} = 0$ and $A_{-1} = 1$. DAGC-V2 updates

$$u_t = r(\tau - E_t) - w \max(0, -A_t), \tag{4}$$
$$c_{t+1} = \Pi_{[L_t, U_t]}(c_t \exp[\gamma \tanh(u_t)]). \tag{5}$$

Algorithm 1 One DAGC-V2 training step

1: Compute raw g_t, n_t, and a_t.
2: Save raw g_t for the next alignment calculation.
3: Apply clipping at the current c_t using Eq. (1).
4: Update exposure E_t, alignment A_t, and norm history.
5: Compute c_{t+1} using Eqs. (4)–(5).
6: Update momentum and parameters using $\widetilde{g}_t$.

Here $L_t = \max(0.1 m_t, 10^{-12})$ and $U_t = \max(10 m_t, 10^{-9})$, where m_t is the lower median of the last 200 raw gradient norms, including n_t. Thus these bounds affect the next step. The frozen settings are $(c_0, \gamma, \beta, \beta_a, r, w, \tau) = (0.5, 0.05, 0.9, 0.9, 0.3, 3, 0.1)$. V1 replaces $r(\tau - E_t)$ with $r(1 - E_t)$; since $E_t \in [0, 1]$, its exposure term cannot itself tighten clipping. V2 can tighten, but this change alone does not imply useful regulation.

The ordering prevents the current norm from selecting its own DAGC threshold. Clipping precedes SGD's coupled weight decay and momentum accumulation. Hence c_t bounds the clipped data gradient, not the final parameter displacement. The method stores a previous full gradient, requiring $O(d)$ additional memory, and a 200-element norm window. Its diagnostic alignment requires an $O(d)$ dot product. We do not describe it as constant-memory control.

3.1. The reference exposure is not a stable target

Consider a stationary, exogenous norm distribution with differentiable CDF F and density f. Remove the alignment term and assume the bounds are inactive. With $x_t = \log c_t$, replace the binary exposure by its conditional mean $q(x_t) = 1 - F(e^{x_t})$. The deterministic mean-field system is

$$E_t = \beta E_{t-1} + (1-\beta) q(x_t), \tag{6}$$
$$x_{t+1} = x_t + \gamma \tanh[r(\tau - E_t)]. \tag{7}$$

Proposition 1. *Suppose $0 \le \beta < 1$, $\gamma r > 0$, and an interior fixed point $c_* > 0$ satisfies $1 - F(c_*) = \tau$ and $f(c_*) > 0$. The mean-field fixed point is linearly unstable.*

Proof. Write $s = c_* f(c_*) > 0$ and $k = \gamma r (1-\beta) s > 0$. For the state (x_t, E_{t-1}), the Jacobian at the fixed point is

$$J = \begin{pmatrix} 1 + k & -\gamma r \beta \\ -(1-\beta) s & \beta \end{pmatrix}. \tag{8}$$

Its characteristic polynomial is $p(\lambda) = \lambda^2 - (1+\beta+k)\lambda + \beta$. Since $p(1) = -k < 0$ and $p(\lambda) \to +\infty$ as $\lambda \to +\infty$, it has a real root greater than one. □

The result includes exposure smoothing; an EMA does not reverse the feedback sign. It does not prove instability of neural-network training: the gradient norm distribution changes with the parameters, the bounds move, and the alignment signal can activate. It does explain why the reference τ should not be interpreted as a regulated clipping frequency. Persistent clipping can instead drive the threshold to its lower bound, where historical norm scale may dominate the resulting behavior.

3.2. What clipping can protect

For momentum SGD without weight decay, $v_t = \mu v_{t-1} + \widetilde{g}_t$. If $c_j \le C$ over k updates and $0 \le \mu < 1$, then

$$\|v_t\| \le \mu^k \|v_{t-k}\| + C\frac{1-\mu^k}{1-\mu}. \tag{9}$$

A smaller sustained threshold bounds new momentum contributions but does not remove an existing buffer. Coupled weight decay adds parameter-dependent terms. This elementary bound motivates a protection hypothesis without establishing convergence, explaining a test-accuracy improvement, or requiring exposure and alignment to be the best threshold signals.

4. Experimental Design

4.1. Questions, conditions, and data separation

The new study asks whether the frozen controller outperforms tuned fixed clipping, a tuned median-scale rule, and calibration-trace replay. Its secondary question is whether removing alignment changes the outcome. The three conditions are FashionMNIST with the existing small CNN at initial learning rate 0.8, and CIFAR-10 with ResNet-20 at 0.1 or 0.4. The latter two are designated lower and higher learning rates; we do not assume either one diverges without clipping.

FashionMNIST (Xiao et al., 2017) uses 55,000 training images and a fixed 5,000-image validation split. CIFAR-10 (Krizhevsky, 2009) uses 45,000 training and 5,000 validation images. Both splits hold out 500 images per class using seed 1729. All methods train on the same remaining examples; validation images never enter training. Only calibration runs evaluate the validation set. Confirmation runs evaluate the complete official 10,000-image test set once, after the last training epoch. They retain the same training partition, preserving the number of steps used to construct replay schedules. No test score is used to choose a new-campaign hyperparameter or checkpoint.

FashionMNIST uses two convolutional layers (32 and 64 channels), max pooling, and a 128-unit hidden classifier, with 421,642 parameters. CIFAR-10 uses the $6n$+2 residual architecture of He et al. (2016), $n = 3$, widths 16/32/64, option-A shortcuts, and batch normalization, with 269,722 parameters. The RGB model is deeper, not larger in parameter count. This is an architectural transfer test, not a reproduction of the original ResNet training recipe.

4.2. Training and calibration budget

Training uses FP32 SGD with momentum 0.9, batch size 256, and no dropped last batch. FashionMNIST runs for 30 epochs with weight decay 10^{-4}; CIFAR-10 runs for 60 epochs with weight decay 5×10^{-4}. A cosine schedule decreases the learning rate to 1% of its initial value. CIFAR-10 uses random padded crops and horizontal flips; FashionMNIST uses normalization only. All settings are fixed across clipping policies within a condition.

Fixed clipping, AGC, AutoClip, and the median rule each receive three candidates and three calibration seeds (2100–2102): nine runs per policy per condition. Their grids are, respectively, $\{0.05, 0.5, 5\}$, $\{0.005, 0.03, 0.18\}$, percentiles $\{1, 10, 50\}$, and $\{0.025, 0.1, 0.4\}$. The winner maximizes mean final validation accuracy, with grid order breaking exact ties. These equal finite budgets do not constitute exhaustive tuning. DAGC retains the setting selected during earlier development, so its total historical tuning budget is not matched to the new baselines.

AGC clips every parameter, grouping matrix and convolution weights by output unit and treating vectors as a single unit, with parameter-norm floor 10^{-3}. AutoClip takes the stated percentile of all raw norms through the current step, following its authors' convention. The median rule clips at $\lambda \,\mathrm{median}(n_{t-200}, \ldots, n_{t-1})$, using available history and initial threshold 0.5. It is causal and uses the same lower-median convention as DAGC. The no-alignment variant sets $w = 0$ without retuning. Replay averages the thresholds actually applied in three full DAGC calibration runs and applies this fixed, step-indexed schedule in confirmation. All eight policies, including no clipping, share a complete training budget.

4.3. Pairing, failures, and inference

Confirmation uses twelve seeds, 3100–3111. Within a seed, initial weights, minibatch order, and augmentations are identical across policies. Private random generators isolate the input stream from diagnostics and controller state. Deterministic CUDA operations are enabled and TF32 is disabled. Epoch checkpoints include model, optimizer, controller, previous raw gradient, and logs; a test verifies that resumed and uninterrupted runs agree exactly. Configuration and source checksums reject stale results.

A run stops on nonfinite loss or gradient norm, or loss above 10^6. A nonfinite final evaluation also fails. Failed runs receive accuracy zero and remain in all aggregates; fi-

Table 1. Final test accuracy (percent), mean ± Student-t 95% interval half-width over twelve paired training seeds. Hyperparameters are selected using the validation set. Failures are retained at zero accuracy.

Policy	FashionMNIST, LR 0.8	CIFAR-10, LR 0.1	CIFAR-10, LR 0.4
No clipping	3.33 ± 3.13	90.67 ± 0.09	90.35 ± 0.28
Tuned fixed	92.60 ± 0.07	90.67 ± 0.09	91.16 ± 0.12
Tuned AGC	92.40 ± 0.05	90.66 ± 0.14	91.20 ± 0.16
Tuned AutoClip	90.46 ± 0.15	90.63 ± 0.12	90.78 ± 0.27
Tuned median scale	92.56 ± 0.12	90.58 ± 0.16	91.08 ± 0.15
DAGC-V2	92.41 ± 0.09	89.50 ± 0.10	90.87 ± 0.11
No alignment	92.26 ± 0.18	89.50 ± 0.10	90.87 ± 0.11
Trace replay	92.39 ± 0.09	89.40 ± 0.15	90.94 ± 0.09

Table 2. Primary paired comparisons in percentage points: DAGC minus the comparator. Intervals are unadjusted 95% paired t intervals; p_H is the exact two-sided sign-flip p-value after Holm adjustment over all nine tests. Win counts exclude ties.

Condition / initial LR	Comparator	Difference	95% interval	Wins	p_H
FashionMNIST / 0.8	Tuned fixed	-0.184	[-0.291, -0.078]	1/12	0.0176
FashionMNIST / 0.8	Tuned median scale	-0.148	[-0.251, -0.044]	2/12	0.0635
FashionMNIST / 0.8	Trace replay	+0.023	[-0.081, +0.126]	5/12	0.7236
CIFAR-10 / 0.1	Tuned fixed	-1.172	[-1.325, -1.019]	0/12	0.0044
CIFAR-10 / 0.1	Tuned median scale	-1.080	[-1.257, -0.903]	0/12	0.0044
CIFAR-10 / 0.1	Trace replay	+0.102	[-0.076, +0.280]	8/12	0.7236
CIFAR-10 / 0.4	Tuned fixed	-0.292	[-0.409, -0.175]	1/12	0.0103
CIFAR-10 / 0.4	Tuned median scale	-0.205	[-0.405, -0.005]	3/12	0.1914
CIFAR-10 / 0.4	Trace replay	-0.068	[-0.200, +0.065]	4/12	0.7236

nite chance-level runs retain their observed score. We do not choose the best epoch, discard failures, or rerun unfavorable seeds. The final audit checks all 117 calibration and 288 confirmation outcomes.

We report Student-t 95% intervals over seeds, including paired intervals for accuracy differences. The primary family comprises DAGC versus fixed, median, and replay in all three conditions: nine two-sided exact sign-flip tests with Holm correction. Sign flipping assumes exchangeability under the null; the t intervals assume the seed-level mean is adequately approximated by that model. Neither addresses uncertainty from sampling a new dataset. A broad superiority claim requires a positive effect and corrected significance in all nine comparisons. Secondary comparisons are descriptive.

A prespecified ± 0.5 percentage-point margin supplies an additional within-comparison equivalence check: the paired 90% interval must lie entirely inside that range. This is not a familywise equivalence claim. Nonsignificance alone is not evidence of equivalence. The original eight-seed plan was amended to twelve during calibration, before any confirmation runs: with eight seeds, the minimum two-sided sign-flip $p = 2/256$ cannot pass the first nine-test Holm threshold $0.05/9$. The audit preserves this amendment and all identical calibration results carried forward from the first execution.

5. Results

5.1. Completeness, failures, and accuracy

All 405 planned outcomes pass the source/configuration, split, seed-coverage, epoch-continuity, calibration-selection, and replay-construction audits. There are 397 full-budget completions and eight predefined scientific failures, with no process errors. All eight failures occur without clipping on FashionMNIST. Its four remaining unclipped runs finish at 10% accuracy; retaining failures at zero gives a mean of 3.33%. Every clipped confirmation run and every CIFAR-10 run completes its budget.

Table 1 reports the full comparison. On FashionMNIST, DAGC reaches 92.41%, versus 92.60% for tuned fixed clipping and 92.56% for the tuned median rule. On CIFAR-10 at LR 0.1, DAGC reaches 89.50%, versus 90.67% for fixed clipping. At LR 0.4, the corresponding scores are 90.87% and 91.16%. AGC attains the highest observed mean at the higher CIFAR rate, 91.20%; differences among non-DAGC baselines are not a prespecified hypothesis family.

The paired DAGC-minus-fixed effects are -0.18, -1.17, and -0.29 points, with Holm-adjusted p-values 0.0176, 0.0044, and 0.0103, respectively (Table 2). Thus the new study reverses the earlier exploratory fixed-baseline advantage. These are different data sizes, schedules, and calibration budgets, so this is a failure of transfer under the new protocol, not a claim that the earlier measurements were numerically false. The prespecified broad-superiority crite-

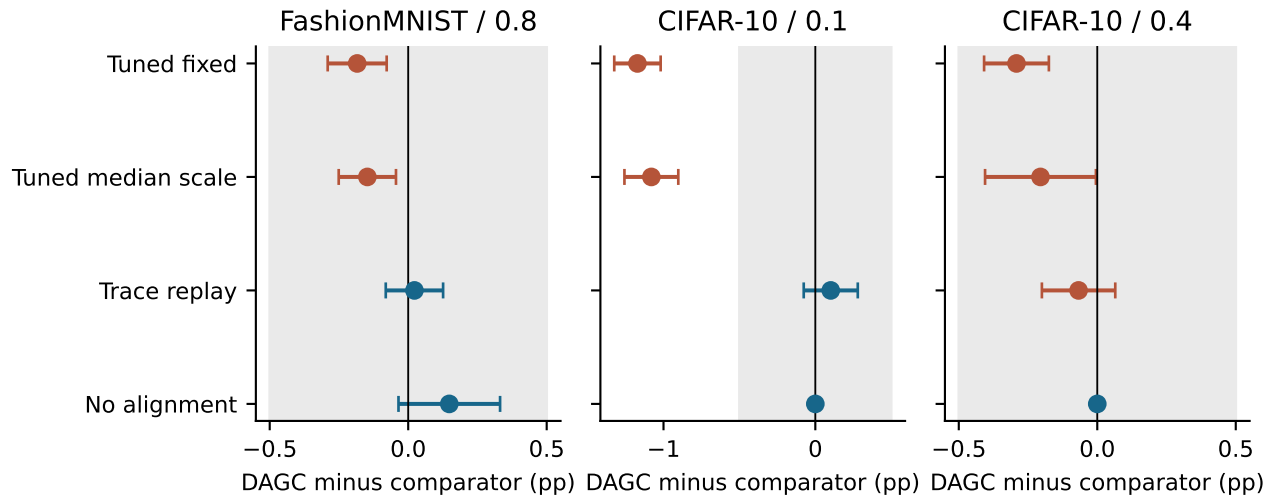


Figure 1. Paired effects of DAGC versus the main controls and the no-alignment variant. Bars show 95% intervals; the shaded region is the prespecified ± 0.5 point margin. Equivalence requires a separate 90% interval check.

rion is not met.

At CIFAR-10 LR 0.1, calibration selects fixed threshold 5.0, and this threshold never binds in confirmation. Its norm traces and coefficients exactly match no clipping. Accordingly, the 1.17-point deficit is also a deficit to the unmodified optimizer. At LR 0.4, DAGC exceeds no clipping by 0.52 points [0.25, 0.79], but stronger performance is achieved by the tuned fixed rule. Avoiding divergence or improving on no clipping is therefore insufficient evidence for the particular DAGC controller.

5.2. Replay and alignment controls

DAGC-minus-replay differences are $+0.02$, $+0.10$, and -0.07 points. Their paired 90% intervals are $[-0.06, 0.11]$, $[-0.04, 0.25]$, and $[-0.18, 0.04]$. Each lies inside the prespecified ± 0.5 point margin. This supports equivalence within each tested condition at that tolerance; it is not a multiplicity-adjusted familywise equivalence result, nor a claim that the methods coincide outside these settings. The corresponding 95% intervals appear in Table 2 and Figure 1.

Removing alignment changes FashionMNIST accuracy by $+0.15$ points in DAGC's favor, with a paired 95% interval $[-0.04, 0.33]$. On both CIFAR-10 conditions, all twelve paired raw-gradient norms, applied thresholds, coefficients, alignment traces, and lower-bound indicators are exactly equal between DAGC and the no-alignment variant. The complete-trace audit shows that negative smoothed alignments do occur; the direction term is therefore not intrinsically inactive. All 3,860 negative-alignment-EMA updates across the two CIFAR conditions are projected to the lower bound. Its effect on the applied threshold is masked by lower-bound projection in these runs. This identifies a limit of the implemented controller, rather than demonstrating that gradient direction never matters.

5.3. The lower bound dominates the controller

DAGC's next-step threshold equals its moving lower bound on 97.18%, 99.07%, and 99.00% of updates, respectively. Its mean clipping intensities are 0.893, 0.900, and 0.899. Figure 2 shows the resulting profiles. These measurements are consistent with the positive-feedback calculation: clipping stays active and the projection frequently determines the threshold.

The validation-selected median coefficient is 0.1 on FashionMNIST, matching DAGC's lower-bound scale, and 0.4 on both CIFAR conditions. The CIFAR median rule therefore retains larger gradients, with intensities near 0.60. DAGC falls below it by 1.08 points at LR 0.1 (Holm $p = 0.0044$) and 0.21 points at LR 0.4 (Holm $p = 0.1914$). The latter comparison does not pass the primary multiplicity criterion. Bound occupancy describes which part of the rule is active; by itself it does not prove that changing only the bound would produce a particular accuracy gain.

5.4. Earlier exploratory evidence and corrected interpretation

The earlier 15-epoch, 5,000-training/2,000-test experiments used different training seeds for development and evaluation, but sampled from the same official test set for both stages. We therefore classify them as exploratory, not independent data-held-out confirmation. Their positive

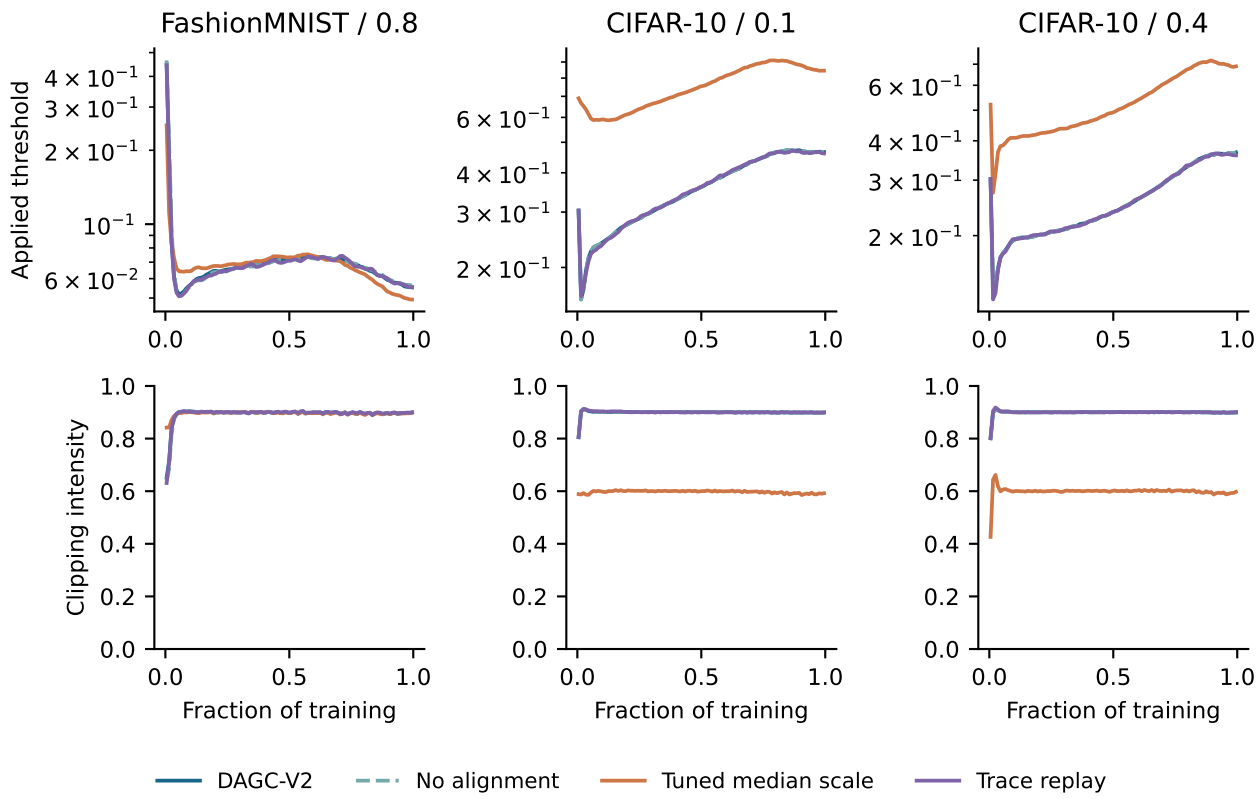


Figure 2. Applied thresholds and clipping intensities averaged over twelve confirmation seeds and 100 bins of training progress. DAGC and no-alignment curves overlap on CIFAR-10. These observational trajectories describe behavior; they do not establish the causal value of an individual feedback signal.

comparisons to fixed clipping motivated the present study; they cannot substitute for it.

The most discriminating legacy experiment already limited the mechanism claim. Over eight paired seeds, DAGC exceeded an intensity-matched fixed rule by 1.18 points (recomputed 95% paired interval [0.62, 1.74]), but its advantage over replay was 0.21 points [−0.43, 0.86]. Removing alignment changed accuracy by 0.16 points [−0.33, 0.66], whereas removing exposure cost 5.47 points [4.04, 6.90]. These results identify the effect of specified ablations on the whole training trajectory. They do not show that reversal predicts future loss deterioration or that online feedback adds value beyond replay.

Independent-seed learning-rate perturbations did not resolve the ambiguity: the DAGC-minus-replay difference was −0.06 [−0.71, 0.59] for a temporary learning-rate doubling and +0.01 [−0.51, 0.52] for a sustained decrease. The uncertainty spans effects beyond the new equivalence margin, so these legacy null results are not equivalence claims. Appendix C reports all controls and corrected intervals.

A development sweep over the reference exposure had previously pooled two gains and three seeds into six apparent replicates. We instead report each gain separately with three seeds. Realized exposure remains at least 0.999 across the grid, while intensity changes. The sweep is consistent with the positive-feedback interpretation, but its small sample and development role preclude precise claims of performance insensitivity.

6. Discussion and Limitations

A controller can influence an optimization trajectory without the proposed feedback signal being necessary for its gains. The replay comparison removes threshold adaptation to the confirmation trajectory; the median comparison retains online magnitude information but removes exposure and alignment; and the no-alignment comparison removes only direction. Their scopes differ, so no single one is a complete mechanistic explanation.

The evidence supports a narrow negative conclusion about the frozen V2 rule: it does not outperform a validation-tuned fixed threshold in any of the three conditions, and its threshold adaptation adds no practically resolved advantage over calibration replay under the stated margin.

The lower bound accounts for nearly all threshold updates, while its directional adjustments are masked by projection on CIFAR-10. Calling these runs successful target tracking or evidence that online directional feedback is necessary would contradict the controls. They instead show how an apparently dynamic controller can behave mostly as a historical-scale clipping rule.

The FashionMNIST and higher-rate CIFAR fixed-baseline gaps are statistically resolved but smaller than 0.5 points. Statistical inferiority can coexist with practical equivalence at a specified tolerance; the larger low-rate CIFAR gap falls outside that tolerance. The finding should guide subsequent research toward identifying conditions where feedback is informative, rather than retuning the present confirmation runs after observing their test scores.

The study remains limited to two image datasets, two small architectures, one optimizer, and fixed learning-rate schedules. Sixty CIFAR epochs are longer than the legacy diagnostics but do not establish asymptotic performance or large-scale training stability. Fixed grids can miss better baseline settings. Historical FashionMNIST test exposure and unequal earlier method development remain limitations despite validation-only selection in the new campaign. Seed intervals are conditional on a fixed data split. Our analytic model freezes the norm distribution; its unstable controller equilibrium is not a theorem about the nonlinear coupled training system. Likewise, lower-bound occupancy and accuracy correlation cannot identify a causal mechanism alone.

7. Conclusion

DAGC-V2 implements positive exposure feedback, not stable target-rate tracking in the stationary mean-field model. A useful empirical claim must therefore survive comparisons with magnitude-only adaptation and a predetermined threshold schedule, in addition to tuned fixed clipping. Our protocol makes those comparisons explicit, preserves failure outcomes, and separates legacy exploration from current validation-only selection. Across all 405 audited runs, the frozen controller fails the broad-superiority criterion and loses to tuned fixed clipping in each confirmation condition. Replay meets the prespecified per-condition equivalence criterion, while the complete CIFAR traces show that lower-bound projection masks the alignment term. Together, these controls limit the evidence for the proposed feedback signals in these tests. These results delimit the method's empirical claim and provide a reproducible example of separating threshold scheduling from evidence for an online feedback mechanism.

Impact Statement

This work examines optimization behavior in small image classifiers. It does not deploy a model or introduce a new dataset. Better-controlled evaluation can reduce wasted experiments and unsupported claims about training stability. A clipping rule is not a guarantee of reliable or safe behavior in downstream applications; broader deployment consequences depend on the model and use.

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A. Reproducibility and Complete Calibration

The machine-readable protocol fixes conditions, hyperparameter grids, calibration and confirmation seeds, and comparison families before confirmation. The source/configuration digest is checked when a run is resumed or reused. The evidence export verifies every run identity, exact confirmation seed sets, completed epoch continuity, and failure accounting. It retains per-epoch training metrics, final evaluation scores, 100-bin diagnostic traces, full replay schedules, and checksums for the full-resolution traces archived with the raw experiment. Tables and figures are generated from the exported seed-level records; summary values are not entered by hand.

Training inputs are cached on the GPU. CIFAR crop offsets and flips use a private generator keyed by training seed and epoch; a separate generator controls the permutation of training examples. No dropout or other random model operation is used. Model initialization uses the training seed. Exact resume tests cover optimizer state, controller state, prior gradient, metrics, and diagnostic traces. The numerical epsilon in scalar clipping is 10^{-12}; AGC's parameter floor is 10^{-3}. CUDA determinism is requested explicitly.

Table 3. Shared implementation details.

Item	FashionMNIST	CIFAR-10
Training / validation / test	55,000 / 5,000 / 10,000	45,000 / 5,000 / 10,000
Epochs / updates	30 / 6,450	60 / 10,560
Model / parameters	SmallCNN / 421,642	ResNet-20 / 269,722
Batch / SGD momentum	256 / 0.9	256 / 0.9
Weight decay	10^{-4}	5×10^{-4}
Mean	0.286	(0.4914, 0.4822, 0.4465)
Standard deviation	0.353	(0.2470, 0.2435, 0.2616)
Augmentation	None	Padded crop and horizontal flip
Evaluation frequency	Final epoch only	Final epoch only

Table 4. Complete calibration grid. Each candidate uses the same three seeds. Entries list candidate: mean validation accuracy (percent). The final column is selected before confirmation.

Condition / initial LR	Policy	Candidate scores	Selected
FashionMNIST / 0.8	Tuned fixed	0.05: 93.04, 0.5: 90.32, 5: 10.00	0.05
FashionMNIST / 0.8	Tuned AGC	0.005: 93.10, 0.03: 63.57, 0.18: 10.00	0.005
FashionMNIST / 0.8	Tuned AutoClip	1: 90.89, 10: 90.33, 50: 36.63	1
FashionMNIST / 0.8	Tuned median scale	0.025: 91.94, 0.1: 93.06, 0.4: 91.17	0.1
CIFAR-10 / 0.1	Tuned fixed	0.05: 77.25, 0.5: 91.16, 5: 91.64	5
CIFAR-10 / 0.1	Tuned AGC	0.005: 84.25, 0.03: 90.31, 0.18: 91.51	0.18
CIFAR-10 / 0.1	Tuned AutoClip	1: 91.34, 10: 91.51, 50: 91.48	10
CIFAR-10 / 0.1	Tuned median scale	0.025: 87.73, 0.1: 90.61, 0.4: 91.17	0.4
CIFAR-10 / 0.4	Tuned fixed	0.05: 88.76, 0.5: 92.01, 5: 91.26	0.5
CIFAR-10 / 0.4	Tuned AGC	0.005: 83.66, 0.03: 91.57, 0.18: 91.45	0.03
CIFAR-10 / 0.4	Tuned AutoClip	1: 91.44, 10: 91.63, 50: 91.44	10
CIFAR-10 / 0.4	Tuned median scale	0.025: 89.85, 0.1: 91.91, 0.4: 92.03	0.4

B. Failures and Clipping Diagnostics

For global clipping, the logged coefficient is the scalar multiplier. AGC instead logs the ratio of the whole clipped-gradient norm to the raw norm; its value does not imply a global scalar intervention, since individual units receive different multipliers. Frequency counts ratios below $1 - 10^{-7}$. Intensity is one minus that ratio. Bound occupancy counts whether the newly computed DAGC threshold equals its lower bound to relative tolerance 10^{-6}. For other policies this diagnostic is zero by construction and does not imply that their thresholds are unconstrained.

Table 5. Failures and mean diagnostics across twelve seeds. f is clipping frequency, I is intensity, and b is lower-bound occupancy.

Condition / initial LR	Policy	Failed	f	I	b
FashionMNIST / 0.8	No clipping	8/12	0.000	0.000	0.000
FashionMNIST / 0.8	Tuned fixed	0/12	1.000	0.947	0.000
FashionMNIST / 0.8	Tuned AGC	0/12	1.000	0.941	0.000
FashionMNIST / 0.8	Tuned AutoClip	0/12	0.954	0.297	0.000
FashionMNIST / 0.8	Tuned median scale	0/12	1.000	0.896	0.000
FashionMNIST / 0.8	DAGC-V2	0/12	1.000	0.893	0.972
FashionMNIST / 0.8	No alignment	0/12	1.000	0.893	0.970
FashionMNIST / 0.8	Trace replay	0/12	1.000	0.894	0.000
CIFAR-10 / 0.1	No clipping	0/12	0.000	0.000	0.000
CIFAR-10 / 0.1	Tuned fixed	0/12	0.000	0.000	0.000
CIFAR-10 / 0.1	Tuned AGC	0/12	0.763	0.008	0.000
CIFAR-10 / 0.1	Tuned AutoClip	0/12	0.960	0.262	0.000
CIFAR-10 / 0.1	Tuned median scale	0/12	1.000	0.598	0.000
CIFAR-10 / 0.1	DAGC-V2	0/12	1.000	0.900	0.991
CIFAR-10 / 0.1	No alignment	0/12	1.000	0.900	0.991
CIFAR-10 / 0.1	Trace replay	0/12	1.000	0.900	0.000
CIFAR-10 / 0.4	No clipping	0/12	0.000	0.000	0.000
CIFAR-10 / 0.4	Tuned fixed	0/12	1.000	0.585	0.000
CIFAR-10 / 0.4	Tuned AGC	0/12	1.000	0.232	0.000
CIFAR-10 / 0.4	Tuned AutoClip	0/12	0.987	0.355	0.000
CIFAR-10 / 0.4	Tuned median scale	0/12	0.999	0.599	0.000
CIFAR-10 / 0.4	DAGC-V2	0/12	1.000	0.899	0.990
CIFAR-10 / 0.4	No alignment	0/12	1.000	0.899	0.990
CIFAR-10 / 0.4	Trace replay	0/12	1.000	0.900	0.000

C. Legacy Exploratory Experiments

These experiments precede the new validation-isolated protocol. Each training seed selects 5,000 official training images and 2,000 official test images; test scores were available during earlier method selection. The CNN is the same SmallCNN. Training uses 15 epochs, 300 updates, constant LR 0.8, batch 256, SGD momentum 0.9, and weight decay 10^{-4}. Calibration seeds 800–802 construct the mean replay trace and select a fixed threshold from $\{0.05, 0.1, 0.2, 0.3, 0.5\}$ by matching mean clipping intensity, not accuracy. That criterion chooses 0.2. Evaluation seeds are 900–907. The new campaign instead tunes fixed clipping for validation accuracy.

Table 6. Legacy mechanism controls, eight paired seeds. All intervals are recomputed Student-t 95% intervals, replacing the original normal approximation. Δ is DAGC minus the listed control, in percentage points.

Policy	Accuracy (%)	Δ	Paired 95% interval
DAGC-V2	86.79 ± 0.66	+0.00	[+0.00, +0.00]
No alignment	86.63 ± 0.93	+0.16	[-0.33, +0.66]
No exposure	81.33 ± 1.24	+5.47	[+4.04, +6.90]
Zero controller gain	80.74 ± 1.04	+6.05	[+4.92, +7.18]
Intensity-matched fixed	85.61 ± 0.93	+1.18	[+0.62, +1.74]
Trace replay	86.58 ± 0.56	+0.21	[-0.43, +0.86]
AutoClip (10th percentile)	79.88 ± 1.74	+6.92	[+5.39, +8.45]
Rate tracking	75.21 ± 1.89	+11.59	[+9.62, +13.56]

"Zero controller gain" sets $\gamma = 0$ but retains projection onto moving norm-dependent bounds. It therefore does not implement a truly fixed threshold. The older label "frozen threshold" obscured that distinction. "No exposure" sets $r = 0$; no alignment sets $w = 0$. AutoClip uses its tenth percentile, without a tuning search. The rate-tracking control reverses the exposure sign and removes the direction term, giving $c^+ = c\exp\{0.05\tanh[0.3(E - 0.1)]\}$ without rolling-median bounds and with the same initial threshold. It is an isolated sign-control experiment, not a replication of a private federated algorithm. These untuned configurations do not establish method-wide superiority over those algorithm families.

Table 7. Legacy learning-rate perturbations, eight paired seeds 1000–1007. DAGC-minus-control accuracy differences in percentage points.

Perturbation	Comparator	Difference	Paired 95% interval
Drop	No alignment	+0.21	[-0.21, +0.64]
Drop	Trace replay	+0.01	[-0.51, +0.52]
Pulse	No alignment	-0.02	[-0.54, +0.50]
Pulse	Trace replay	-0.06	[-0.71, +0.59]

The pulse changes LR from 0.8 to 1.6 at zero-based step 100 and back to 0.8 at step 160. The drop changes LR from 0.8 to 0.2 at step 100 and keeps it there. The replay remains the unperturbed calibration trace. Neither condition recalibrates or resets the optimizer.

Legacy comparisons labeled ZClip or AdaGC are not used as evidence here: the correspondence of their local implementations and modified warmup to the authors' algorithms has not been established. Their raw artifacts remain in the project history, but calling those settings "published defaults" would not be justified. The present baseline claims are limited to the explicitly defined, calibrated clipping rules in the new protocol.

Table 8. Legacy development sweep, three seeds (200–202) per gain/reference combination. Gains are not pooled as independent replicates. Intervals reflect only within-setting seed variation and are descriptive, not selection-adjusted.

Gain γ	Reference τ	Accuracy (%), 95% interval	Exposure	Intensity
0.05	0.10	86.83 ± 3.50	1.000	0.799
0.05	0.20	86.20 ± 2.69	1.000	0.779
0.05	0.35	86.08 ± 2.59	1.000	0.749
0.05	0.50	85.87 ± 3.39	1.000	0.716
0.05	0.70	85.85 ± 4.00	0.999	0.633
0.10	0.10	86.52 ± 3.73	1.000	0.850
0.10	0.20	86.47 ± 5.54	1.000	0.840
0.10	0.35	86.53 ± 2.76	1.000	0.827
0.10	0.50	86.12 ± 2.87	1.000	0.801
0.10	0.70	86.62 ± 2.08	1.000	0.762